

Paolo L. Sinopoli

Education

Massachusetts Institute of Technology

Biology Ph.D. Student

NSF Graduate Fellow

Start: September 2026

The Ohio State University

Bachelor of Science in Molecular Genetics (*with Honors Research Distinction*)

GPA: 4.00/4.00

Provost Scholarship

August 2022 – May 2026

Research Positions

Massachusetts Institute of Technology, Cambridge, MA

Research Intern (Advisor: Dr. Anders Sejr Hansen)

June 2026 – September 2026

The Ohio State University, Columbus, OH

Undergraduate Researcher (Advisor: Dr. Anita Hopper)

Project: Free introns of tRNAs (fitRNAs) as novel noncoding regulatory RNAs

October 2022 – May 2026

Scripps Research, San Diego, CA

Research Intern (Advisors: Dr. Megan Ken & Dr. Jamie Williamson)

Project: Observing the assembly of HIV-1 Gag proteins on a supported lipid bilayer

May 2025 – August 2025

National Cancer Institute, Frederick, MD

Research Intern (Advisor: Dr. Sandra Wolin)

Project: Investigation of Y RNA-dependent Ro60 protein interactions in *Salmonella Typhimurium*

May 2024 – August 2024

Research Fellowships and Awards

National Science Foundation Graduate Research Fellowship – \$159,000

– *Proposal title: “Nanoscale architecture and dynamics of mammalian Pol III transcription factories”*

April 2026

Center for RNA Biology Trainee Spotlight

March 2026

Semi-Finalist: Hertz Foundation Graduate Fellowship

November 2025

RNA Society Annual Meeting Oral Presentation Prize

June 2025

Undergraduate Research Scholarship

April 2023, -24, -25

– *Awarded three years consecutively totalling \$16,900*

Invited Speaker at RNA Society Annual Meeting

April 2025

– *Only undergraduate speaker; >1,000 attendees*

Barry M. Goldwater Foundation Scholarship – \$7,500

March 2025

RNA Society Research Presentation Fellowship – \$1,000

February 2025

Invited Speaker at Rustbelt RNA Meeting Undergraduate Symposium

October 2024

Eleanor Ruffing McMahon Conference Travel Award

October 2024

NCI Cancer Research Training Award – \$8,900

May 2024

Center for RNA Biology Symposium Best Poster Award

May 2024

STEP Fellowship – \$2,000

May 2024

Pelotonia Foundation Undergraduate Cancer Research Fellowship – \$12,000

April 2024

– *One of 13 awarded in 2024*

Thomas J. Byers Memorial Research Scholarship

March 2024

– *Declined due to mutually exclusive funding*

Rustbelt RNA Meeting Poster Excellence Award
Department of Molecular Genetics Travel Funding
Jim Hopper Memorial Undergraduate Research Award
– Declined due to mutually exclusive funding

October 2023
September 2023
April 2023

Publications

1. Nostramo RT, **Sinopoli PL**, Bao A, Metcalf S, Peltier LM, Hopper AK. (2025). Free introns of tRNAs as novel complementarity-dependent regulators of gene expression. **Molecular Cell**. PMID: 39938518.
Featured article of the issue. Highlighted by: Cell Press, OSU News, Science Direct, and others.
2. Metcalf S, Nostramo RT, Bao A, Clark K, **Sinopoli PL**, Hopper AK (2026). Many paths to destruction: Family-specific turnover and stress-responses for tRNA introns. **PNAS**. PMID: 41961856. Preprint on BioRxiv.

Oral Presentations

RNA Society Annual Meeting, San Diego, CA June 2025
Invited Speaker – “Free tRNA introns act as novel complementarity-dependent noncoding regulatory RNAs”
Won RNA Society Oral Presentation Prize (1 awarded)

Rustbelt RNA Meeting Undergraduate Symposium, Newark, OH October 2024
Invited Speaker – “Free tRNA introns act as novel noncoding regulatory RNAs”

Select Poster Presentations

Denman Undergraduate Research Forum, Columbus, OH March 2026
Poster – “tRNA introns: Ancient RNAs that hunt genetic messages”

Scripps Research Summer Poster Session, La Jolla, CA August 2025
Poster – “Observing the assembly of HIV-1 Gag proteins on a supported lipid bilayer”

NIH Poster Day, Bethesda, MD August 2024
Poster – “Investigation of Y RNA-dependent Ro60 protein interactions in *Salmonella Typhimurium*”

The Center for RNA Biology Symposium, Columbus, OH May 2024
Poster – “fitRNAs: A new class of regulatory small noncoding RNAs”
Won Best Poster Award (5 awarded from 64 posters)

Rustbelt RNA Meeting, East Lansing, MI October 2023
Poster – “tRNA introns act as novel noncoding regulatory RNAs”
Won Poster Excellence Award (4 awarded from 169 posters)

Department of Molecular Genetics Open House, Columbus, OH September 2023
Poster – “The biological significance of tRNA introns”

Science Outreach, Service, and Teaching

Introduction to Life Sciences: Research Opportunities and Career Options August 2024 – October 2025
Undergraduate Panelist

Phenomenon Design Summit, La Jolla, CA July 2025
Speaker – “Biophysics and Structural Biology: the Assembly of Biomolecules” (tailored to K-12 teachers)
– Topic selected by science teachers for adaptation to high-school lesson plans
– Helped develop lesson plan “Virus Gone Viral” that is taught in San Diego High Schools

DNA Workshops in Columbus Public Schools January 2025 – May 2025

Team Leader – Molecular Biology & Gel Electrophoresis Workshops

Village Book Builders

September 2023 – May 2025

Virtual science tutor for high school students in Kadzkalowa, Malawi

– *Helped fundraise to re-open the local library after closure due to financial obstacles*

TEDxOhioStateUniversity

August 2023 – May 2024

Committee Member

Inaugural Schoenberg Public Lectureship, Columbus, OH

November 2023

Public Outreach Poster – “Unconventional mRNA silencing in baker’s yeast: junk to gold?”
